WASTEWISE SCAN · STEM CHECK

WasteWise Scan STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.8 upgrade) of option A, the current design, and option B; window crosstalk (S12) failed and was fixed (F1); the independent review refuted S5, S7, S18 and S36, fixed by F2 to F4 under Amish's standing instruction of 2026-10-23; assumption-driven rows relabelled plausible; review recorded

WasteWise Scan STEM check

Verdict: the working principle of the scanner (near-infrared reflectance at discrete bands, read by one InGaAs photodiode and classified on band ratios) sits within known science, and no row fails after four fixes made in this session under Amish's standing instruction of 2026-10-23 (fix rule; WSC-DEC-001 F1 to F4). The first run failed one detail: the open-air reading cannot hold the window crosstalk under 0.5 % of the signal (S12, 10 %). The independent review then refuted two "within" verdicts and one "plausible" one: with the dark gap taken from the real reading time, a backed clear item carried 0.73 % against 0.5 % (S5, S7); at the peak of a half-sine drop pulse the cell strap reached 59 MPa against about 50 MPa (S18); and option B's lamps sat 0.2 mm from PETG (S36). The fixes are a baffle tube through the glass with a separate detector window (F1), a scan of four averaged cycles (F2; 0.64 s), an 11 x 10 mm strap (F3) and aluminium lamp sleeves in option B (F4). The review also showed that many rows rest on assumed inputs (cap, rim and tube leaks, the ambient change rate, screen reflectance, crush distance, LED drift coefficients, the IEC 62471 limits as understood without the purchased text), so those rows are now **plausible** until measured. Of the 23 claims of option A, **12 are within known science and 11 are plausible**; of the 7 claims of option B, 4 are within and 3 plausible. S12, S17, S18, S19, S22 and S23 cannot be closed by calculation (they need a built head, a drop test, measured LED coefficients, users or a labelled item set at TRL 4), so `stem_check` is **open**. The kit's verdict label reads "to confirm at TRL 3"; for these rows confirmation is TRL 4 work. Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which runs the calculation note's script `docs/04-calcs/sizing.py` so the two never disagree; results in `docs/04-calcs/stem_results.csv`.

Claims checked

Table 1. Option A (the current design): claims, governing science and verdicts.

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Common resins absorb at C-H overtone and combination bands near 1,150 to 1,250 nm, 1,370 to 1,420 nm and 1,650 to 1,765 nm, at positions that differ between PET, HDPE, PP, PS and PVC	Chemistry	Analytical	Near-infrared spectroscopy of polymers (Neo et al. 2022, review); open Plastic Scanner prior work	Band positions used in WSC-CAL-001 B4; discrete-band identification shown by Plastic Scanner	Within known science	Open
S2	A standard InGaAs photodiode responds from about 900 to 1,700 nm; the 1,650 nm band is seen with its centroid at 1628 nm and 31 % of its signal at or above 1,650 nm	Physics	Optics and radiation	InGaAs band gap cutoff near 1,700 nm at room temperature; responsivity curve of WSC-CAL-001 Table 1	Band integral of a Gaussian LED times the responsivity [B1, B2]: centroid 1628 nm	Within known science	WSC-CAL-001 B2
S3	Signal-to-noise ratio on the weakest band in shade (clear item, 8 conversions in 4 cycles) is at least 600, enough to resolve 0.5 % reflectance differences at 3 sigma (fix F2)	Physics	Electromagnetism and circuits	Shot noise 2qIB and Johnson noise 4kTB/R of the transimpedance stage plus ADC noise	642.503 at least 600	Within known science	WSC-CAL-001 C4
S4	Sunlight through a clear item at 100 klx	Physics	Optics and radiation	Diffuse transmission and view factor of the	1.066 V at most 2.048 V		

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	keeps the 47 kOhm amplifier inside the 2.048 V ADC range			36 mm rim; Ohm's law		Within known science	WSC-CAL-001 D1, D2
S5	Ambient error on a clear item backed with the black cap is under 0.5 % of the 1,650 nm signal	Physics	Optics and radiation	Interleaved dark subtraction; 10^-3 of the through-light left by the cap	0.208 % at most 0.5 %; rests on the assumed cap leak (10^-3) and ambient change (1 % per 10 ms), to measure at TRL 4	**Plausible, to confirm at TRL 3**	WSC-CAL-001 D6 (fix F2); cap opacity at TRL 4
S6	Ambient error on an opaque item is under 0.5 % of the 1,650 nm signal	Physics	Optics and radiation	Rim leak of 10^-3 of the clear-item case	0.075 % at most 0.5 %; rests on the assumed rim leak; the black TPU shroud's near-infrared opacity is to confirm	**Plausible, to confirm at TRL 3**	WSC-CAL-001 D5; shroud opacity at TRL 4
S7	The dark-level rule refuses an unbacked clear item in sun (dark level above 0.71 times the weakest band)	Engineering and technology	Sensing resolution	Error budget: 0.5 % limit over the drift of 1 % per 10 ms in a 7 ms gap	936 times the band reading at least 1.241 times the band reading	Within known science	WSC-CAL-001 D7
S8	Button to result in 1.5 s or less	Engineering and technology	Control	Timing budget: 4 cycles of 17 readings of 2 ms settling plus 2 conversions at 330 samples/s, plus classification and display (fix F2)	0.64 s at most 1.5 s	Within known science	WSC-CAL-001 E2
S9	One 3,000 mAh cell lasts an 8 h shift at 4 scans per minute	Engineering and technology	Power and link budgets	Energy balance: 80 % of 10.8 Wh over the average draw; an 85 % efficient 3.3 V	21.9 h at least 8 h	Within known science	WSC-CAL-001 F2

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				converter still leaves 18.6 h			
S10	A protected 18650 Li-ion cell of 3,000 mAh at 3.6 V stores about 10.8 Wh and is charged at 0 to 45 C only	Chemistry	Electrochemistry and batteries	Cell capacity times nominal voltage; IEC 62133-2 (IS 16046 Part 2) cell safety practice	3.0 Ah x 3.6 V = 10.8 Wh; the 45 C charge limit is the usual Li-ion maker limit, to confirm on the chosen cell's datasheet	Within known science	WSC-CAL-001 F2; WSC-REG-001 R3
S11	The 8 MB log partition holds at least 10,000 scans of 64 bytes	Mathematics	Units and dimensional analysis	8 x 1,048,576 B / 64 B; a margin of 13 times covers file system overhead	131072 scans at least 10,000 scans	Within known science	WSC-CAL-001 G1
S12	LED light reflected inside the window reaches the detector at under 0.5 % of the signal after the open-air reading, with the window offset changing 10 % between readings (PR-017)	Physics	Optics and radiation	Fresnel reflection (4 % per uncoated surface); geometric optics of the baffle through the glass (fix F1); leak through the tube wall	0.0026 % at most 0.5 %; 10 % before fix F1 (Fails, fix proposed); after it the glass path is closed by geometry and the wall leak rests on assumed values (wall opacity 2e-02 or better needed), to confirm on a coupon and a built head at TRL 4	**Plausible, to confirm at TRL 3**	Open
S13	Carbon-black items reflect about 4 % in the near infrared and read 'unknown' below the 0.08 reflectance threshold	Chemistry	Materials and polymer chemistry	Carbon black absorbs across the near infrared (WRAP, black plastic packaging)	Black item returns 8 % of a typical item; separated with SNR over 257 [C6]; an unbacked clear item (0.10) sits close to the threshold and	Within known science	WSC-CAL-001 C6

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					may read 'unknown', which is the safe side		
S14	Corneal irradiance of the brightest LED at 200 mm is inside the IEC 62471 exempt group (100 W/m2)	Physics	Optics and radiation	Radiant intensity over distance squared; IEC 62471 exempt limit for infrared radiation to the eye	3.93 W/m2 at most 100 W/m2; limit values as understood from IEC 62471, standard not purchased; 241 W/m2 at the rim if an LED sticks on, so the hardware on-time limit is required	**Plausible, to confirm at TRL 3**	WSC-CAL-001 H2; limits to confirm against the standard text
S15	Weighted retinal thermal radiance of the 850 nm LED is under the IEC 62471 exempt limit (weak visual stimulus)	Biology	Human physiology and biomechanics	Retinal thermal hazard of near-infrared sources, IEC 62471	3.8 % of the limit at most 100 % of the limit; limit values as understood from IEC 62471, standard not purchased	**Plausible, to confirm at TRL 3**	WSC-CAL-001 H3; limits to confirm against the standard text
S16	The sun hood gives at least 3:1 display contrast on the part of the screen it shades in 100 klx sun	Physics	Optics and radiation	Contrast ratio with diffuse screen reflection of sky light (15 klx)	5.2 :1 at least 3 :1; screen reflectance of 2 % is assumed; a glossy screen reflects sky light specularly (reviewer: about 2:1), so PR-007 stays at risk	**Plausible, to confirm at TRL 3**	WSC-CAL-001 I2 to I4; outdoor check at TRL 4
S17	A 1.2 m drop on a shell corner with about 1 mm of crush decelerates	Physics	Mechanics and statics	Work-energy: deceleration = drop height / stopping distance	claimed 1200 g, estimate 1.2e+03 g (ratio 1.00); mean deceleration	**Plausible, to confirm at TRL 3**	Open

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	the scanner at about 1,200 g				from an assumed 1 mm crush; the peak of a half-sine pulse is 1.57 times higher (used in S18); crush to measure in the drop test at TRL 4		
S18	The printed PETG cell strap stays below the bending strength of printed PETG under the corner-drop cell load	Physics	Strength of materials	Simply supported beam, central peak load: stress = F L / (4 Z); printed PETG about 50 MPa along the layers	34.4 MPa at most 50 MPa; 11 x 10 mm strap at the half-sine peak (fix F3); the 10 x 8 mm strap reached 59 MPa (Fails, fix proposed); rests on the assumed crush (S17), drop test at TRL 4	**Plausible, to confirm at TRL 3**	Open
S19	With LED output falling 0.4 %/C, the 5 % drift warning trips after about 12.5 C of change	Engineering and technology	Sensing resolution	Linear temperature coefficient (typical for SWIR LEDs; assumed in WSC-CAL-001 Table 1)	0.05 / 0.004 = 12.5 C [J3]; the coefficient is assumed, SWIR LEDs may drift more and shift in wavelength on the steep 1,650 nm detector edge, and a uniform drift cancels in the band ratios while a band-to-band difference does not	**Plausible, to confirm at TRL 3**	Open
S20	PVC in a PET stream spoils the batch because PVC degrades well	Chemistry	Materials and polymer chemistry	Thermal dehydrochlorination of PVC releases HCl; PET is melt-	Qualitative claim in WSC-PRB-001; the scanner only reports PVC, it	Within known science	Open

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	below PET melt-processing temperatures			processed hot (polymer handbooks)	does not depend on the temperatures		
S21	Sunlight at about 100 klx carries about 300 W/m2 between 900 and 1,700 nm	Mathematics	Orders of magnitude	About 1,000 W/m2 global irradiance at 100 to 120 klx, times the share of a 5,778 K black body between 900 and 1,700 nm (Planck's law, computed here; atmospheric water bands lower it a little)	claimed 300 W/m2, estimate 253 W/m2 (ratio 1.19)	Within known science	WSC-CAL-001 D1
S22	The 34 mm body thickness sits in the 32 to 51 mm handle range for a hand grip; the 62 mm width is held like a phone for a light pressing task	Biology	Ergonomics	NIOSH 2004-164 handle diameter for single-handle power tools	34 mm at least 32 mm	**Plausible, to confirm at TRL 3**	Open
S23	Eight discrete bands classify the five resins at 95 % or better on clean, non-black items (PR-001)	Mathematics	Statistics and uncertainty	Paper band study: linear discriminant analysis on log band ratios of approximate reference spectra	Not shown at the assumed scatter: 89.5 % at 1.5 % band scatter, 96.7 % at 1.0 %; PE against PP 93 % [B4 to B7]; approximate spectra, train and test from one generator, no 'other' class; needs real item scatter of about 1 % or less; settled only by a labelled item set at TRL 4	**Plausible, to confirm at TRL 3**	Open

Table 2. Option B (tunable-filter spectrometer, WSC-OPT-001): claims, governing science and verdicts. Numbers are first-order estimates in `stem_check.py` (B_ASSUMPTIONS).

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S30	Option B: a 2,400 K tungsten filament radiates at least 10 % of its output between 1,550 and 1,850 nm	Physics	Thermodynamics	Planck's law, numerical integral	12.7 % at least 10 %	Within known science	Open
S31	Option B: the MEMS-FPI sensor's 1,550 to 1,850 nm range covers the PE and PP first-overtone bands from 1,680 to 1,765 nm	Physics	Optics and radiation	Hamamatsu C13272 series range 1,550 to 1,850 nm (Photonics Online datasheet notice)	1,680 to 1,765 nm lies inside 1,550 to 1,850 nm	Within known science	Open
S32	Option B: signal-to-noise ratio of at least 200 per 20 nm step on a clear item in shade	Physics	Electromagnetism and circuits	Lamp irradiance per spectral slice, Lambertian item, shot and Johnson noise	9776 at least 200	Within known science	Open
S33	Option B: button to result in about 0.74 s, under 1.5 s	Engineering and technology	Control	Lamp warm-up plus a 100-point filter sweep at an assumed 5 ms per step	0.74 s; the filter settling time per step is to assess from the sensor datasheet	**Plausible, to confirm at TRL 3**	Open
S34	Option B: the five resins can be told apart from 1,550 to 1,850 nm alone	Chemistry	Analytical	First-overtone C-H bands of PE and PP and the aromatic bands of PET and PS lie in this range; PVC's features are weaker	Plausible from the band positions of WSC-CAL-001 B4; PVC and PET need a labelled item set	**Plausible, to confirm at TRL 3**	Open
S35	Option B: one 3,000 mAh cell lasts an 8	Engineering and technology	Power and link budgets	Energy balance with the lamp energy per scan	19.3 h at least 8 h	Within known science	Open

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	h shift with the lamps						
S36	Option B: the lamps stay clear of softening the black PETG head block (glass transition about 80 C)	Physics	Heat transfer	Average lamp power 0.057 W; bulb glass reaches well above 80 C during each scan	Fails, fix proposed at first (bulb 0.2 mm from PETG, independent review); fix F4 in the option B model: each bulb sits in an aluminium sleeve with a 0.2 mm air gap, which spreads the heat to the block; sleeve temperature to confirm	**Plausible, to confirm at TRL 3**	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**.

The failed details and their fixes

S12, window crosstalk (fix F1). When the check was first run, S12 read 10 % of the signal against the 0.5 % limit: LED light reflected by the item-side surface of the single 25 mm window reached the photodiode through the glass under the baffle at about the level of the item signal (WSC-CAL-001 C5), and the open-air reading removes only the fixed part of it (C7). The verdict was **Fails, fix proposed**. Amish had accepted on 2026-09-25 the fallback of a baffle extended through the glass with separate LED and detector windows (option (b) of WSC-DDR-003 D11), to be made if the offset proved unstable; the STEM check shows on paper that it is, so under the fix rule the fallback is now in the design, model, drawings, CAL, BOM, build plan and PRS (PR-017). Every ray that reaches the photodiode must now enter the detector window from the item side, so the glass path is closed by geometry. The first estimate of what is left (0.010 %) was scaled from the Fresnel path; the reviewer showed that was the wrong path, so C8 now estimates the light that hits the tube wall directly and leaks through it or round the gap (sealed with black RTV): about 0.05 % of the signal, which the open-air reading removes, and 0.003 % for a 5 % change between readings. The wall needs an opacity of 2×10^{-2} or better; carbon-black filament should give far more, but FDM layer gaps can leak, so S12 is **plausible** until a coupon and a built head are measured at TRL 4. Options weighed:

Option	Crosstalk residual	Cost and build	Verdict
Keep the open-air reading only	About 10 % of signal with a 10 % window change (C7)	None	Fails the 0.5 % limit (not chosen)
Anti-reflection coated window	About 1 % (coating cuts the 4 % surface reflection to about 0.5 %, an assumed typical value)	Coated window, price to assess	Still over 0.5 %
Baffle through the glass, separate detector window, gap sealed (chosen)	Glass path closed; wall leak 0.003 % on assumed values (C8)	One 8.4 mm cored hole, a 5.8 mm disc and an RTV bead; line prices unchanged, small disc to assess	Plausible, to confirm at TRL 4

S5 and S7, dark gap (fix F2). The reviewer found the 7 ms gap between a band reading and its dark readings inconsistent with a 14.1 ms reading. Taking the gap as one reading, the backed clear item carried 0.73 % against 0.5 %. The scan now runs four cycles of 17 readings of two conversions each (8.1 ms) and averages them, treating ambient changes about 140 ms apart as independent (an assumption): 0.21 % on a backed clear item, prompt threshold 1.24, 0.64 s per scan. The same change raises the weakest-band signal-to-noise ratio to 643, which meets the 3-sigma criterion of 600 the reviewer asked for in S3.

S18, cell strap (fix F3). At the half-sine peak (1.57 times the mean deceleration) the 10 x 8 mm strap reached 59 MPa against about 50 MPa. It is now 11 x 10 mm with counterbored screw heads: 34 MPa, margin 1.45, still resting on the assumed 1 mm crush (S17).

S36, option B lamp heat (fix F4). The reviewer judged a bulb 0.2 mm from PETG a likely failure; option B's model now puts each lamp in an aluminium sleeve.

The kit 1.9 model also showed that the head block had no light path below the LEDs (each LED face sat on about 1.2 mm of printed block); a 4 mm exit channel from each LED face to the window is now modelled. That is a modelling correction, not a change of what the design does; it is listed with F1 in WSC-DEC-001.

Fields not relevant

- **Biology, microbiology, toxicology and epidemiology:** the scanner makes no biological claim and contains no living or biological material. Contact with contaminated waste is a handling hazard covered by the safety notes in WSC-PRC-001 and the hazard analysis WSC-RSK-001 (gloves; never use the result to judge whether waste is safe).
- **Physics, fluid mechanics, acoustics and vibration, nuclear and ionising radiation:** no fluid flow, no acoustic function and no ionising source. Vibration enters only as the drop case (S17, S18).
- **Chemistry, inorganic and environmental and atmospheric:** no reaction is used; the only chemistry claims are the polymer bands (S1, S13, S20) and the cell (S10).

Concept options

- **Option A, eight-band LED scanner (current design):** no failed row after fixes F1 to F3; 11 rows plausible. Kept as the repo's design; the concept selection is with Amish (WSC-OPT-001, within 10 %).
- **Option B, tunable-filter spectrometer:** no failed row. S30, S31, S32 and S35 within known science; S33 (sweep time), S34 (five resins from 1,550 to 1,850 nm alone) and S36 (lamp heat, after fix F4) plausible and open. Option B would need the same four-cycle scan for sunlight, which is not yet counted in its 0.74 s. Not dropped; scored in WSC-OPT-001.

Independent review

A second agent, run headless (`claude -p`) with read access only to this document, the calculation note WSC-CAL-001, the product requirements WSC-PRS-001 and `sizing.py` , and none of the author's reasoning, was asked to refute the headline results and every "within known science" verdict (2026-10-24). It reproduced the arithmetic of C1 to C4, D1 to D3, E2, F2, G1, H2, I1, I2, J1 and J4 by hand, and found the weaknesses in the inputs and the labels. What it found and what changed:

#	Finding (reviewer's severity)	What changed
1	S12: the 0.010 % after fix F1 was scaled from the Fresnel path; direct LED light on the tube is stronger; wall leak does not see window dirt; the 0.2 mm gap and FDM layer gaps can leak (major)	C8 rebuilt on the direct-flux path with the opacity needed (2 x 10^{-2}); gap sealed with black RTV (build plan step 4, BOM line 16); S12 and PR-017 now plausible and to verify at TRL 4 with a wall coupon and dirty-window readings
2	The cored hole is not "hand tools only" and adds a seal path and a breakage risk (minor)	PR-016 now says "hand tools and a drill"; the RTV bead seals the gap; glass-drilling hazard H6 in WSC-RSK-001 and safety glasses in the build plan
3	S5, S7: the 7 ms dark gap is inconsistent with a 14.1 ms reading; taken from the timing, the backed clear item fails at 0.73 % (major)	Fix F2: four averaged cycles of 8.1 ms readings; 0.21 %, prompt at 1.24, scan 0.64 s
4	S5, S6, S12 rest on assumed leaks and ambient change; TPU shroud opacity unstated (major, labelling)	Relabelled plausible; shroud opacity added as a measurement at TRL 4
5	S3: SNR 200 is only 1 sigma at 0.5 % (major)	Criterion now 600 (3 sigma); met at 643 after fix F2
6	S23: the check (89.5 %) does not support 95 %; synthetic spectra, one generator, no "other" class (major)	Check text now says "not shown at the assumed scatter" and names the scatter needed (1 % or less); stays plausible and open, PR-001 at risk
7	S19: assumed coefficient; wavelength shift on the 1,650 nm edge; 5 % warning loose against 1.5 % scatter (major)	Relabelled plausible; per-band 1.5 % warning proposed to Amish (WSC-DEC-001 open decision 3), since it changes the user's routine
8	S17, S18: the check repeats the claim; the peak of a half-sine pulse is 1.57 times the mean, which puts the strap at 59 MPa (major)	Fix F3: 11 x 10 mm strap, 34 MPa at the peak; S17 and S18 plausible pending the drop test; PR-015 now to verify at TRL 4
9	S16: 2 % screen reflectance not from a datasheet; specular sky reflection gives about 2:1 (minor)	Relabelled plausible; PR-007 stays at risk
10	S14, S15: IEC 62471 limits from memory; 241 W/m2 at the rim if stuck on; beam half-angle 10 deg in H and 15 deg in C5 (minor)	Relabelled plausible; hardware on-time limit stays required (WSC-REG-001 R2). The two half-angles are each the worst case for their check (narrow for eye safety, wide for crosstalk) and are kept
11	S13: clear items near the 0.08 threshold may read "unknown" (minor)	Noted in the row; that is the safe side, and clear items in sun are read backed (effective 0.18)
12	S21 circular; S11 file system overhead; S10 names no cell; S9 has no converter efficiency (note)	S21 now uses Planck's law for the solar share; S11 notes a 13 times margin; S10 says "to confirm on the chosen cell's datasheet"; S9 notes 18.6 h with an 85 % converter
13	S22: NIOSH figure is for a round handle and a US population (minor)	Kept plausible and open for co-design
14	S36: bulb glass 0.2 mm from PETG is likely a failure (minor)	Fix F4: aluminium lamp sleeves in option B; stays plausible
15	Process: review not yet recorded; self-graded closure of S12; "TRL 3" label for rows that close at TRL 4; PR-015 and PR-017 counted as met (minor)	This section; S12 reopened as plausible; the TRL note in the verdict paragraph; PR-015 and PR-017 now to verify at TRL 4

The reviewer's overall verdict was: "the physics is mostly sound, but S12 is not closed, S5 and S7 depend on an inconsistent timing gap, and several 'Within known science' rows are really assumption-driven 'Plausible' rows." All three points are addressed above; none changes the working principle, the pitch or the scope.

Sources

- Neo, E. R. K., et al. 2022. "A Review on Chemometric Techniques with Infrared, Raman and Laser-Induced Breakdown Spectroscopy for Sorting Plastic Waste in the Recycling Industry." *Resources, Conservation and Recycling* 180: 106217. https://doi.org/10.1016/j.resconrec.2022.106217.
- Plastic Scanner project. https://plasticscanner.com/ and https://github.com/Plastic-Scanner; Hackaday, 2021-12-13, https://hackaday.com/2021/12/13/an-open-source-detector-for-identifying-plastics/.
- WRAP. Recyclability of black plastic packaging. https://www.wrap.ngo/resources/report/recyclability-black-plastic-packaging.
- IEC 62471:2006, *Photobiological safety of lamps and lamp systems* (exempt group limits as used in WSC-CAL-001 H; standard not purchased, values to confirm against the text); BAuA report F 2115, *Photobiological Safety of Light Emitting Diodes (LEDs)*, 2013, https://baua.de/EN/Service/Publications/Report/F2115, checked 2026-10-25 (risk groups of EN 62471).
- IS 16046 (Part 2):2018, equivalent to IEC 62133-2:2017, safety requirements for portable sealed secondary lithium cells. BIS, https://www.bis.gov.in/wp-content/uploads/2026/03/UTRF_16046_2_2018.pdf, checked 2026-10-25.
- NIOSH. 2004. *Easy Ergonomics: A Guide to Selecting Non-Powered Hand Tools*. DHHS (NIOSH) Publication No. 2004-164, p. 8 (handle diameter 1 1/4 to 2 in for power tasks). https://cdc.gov/niosh/docs/2004-164/pdfs/2004-164.pdf, checked 2026-10-25.
- ASTM G173, reference solar spectral irradiance (share of global irradiance between 900 and 1,700 nm, rounded to 30 %).
- Hamamatsu C13272 series MEMS-FPI spectrum sensor, range 1,550 to 1,850 nm, tunable filter and InGaAs PIN photodiode in one package. Photonics Online datasheet notice, https://www.photonicsonline.com/doc/ultra-small-mems-fpi-spectrum-sensor-c-datasheet-0001, and Photonics.com, https://www.photonics.com/Product.aspx?PRID=61532 (TO-5 package), both checked 2026-10-25.
- Planck's law, shot noise and Johnson noise as in any optics and electronics textbook; the calculation note WSC-CAL-001 for every number of option A.